

PSYC2016: Attention lecture 2

2024-03-09

PDF version of these slides

What are you unclear on from today's lecture?

Bottleneck ...	bottleneck problem ...	bottlenecks ...
the bottleneck thing ...	what actually is a bottleneck ...	bottlenecking??? ...
bottlenecks. what is a bottleneck? mental? physical? no definition ...	what is a bottleneck? Is it just a gap between perception and processing? ...	Processing before bottleneck ...
Are physical and mental bottlenecks referring to the same thing or are they 2 different concepts ...	the idea of the central bottleneck did not feel properly defined as a concept, more just felt like a buzz word to refer to any sort of cognition ...	so researchers don't know why there is a bottle neck effect then? ...

I've updated the textbook with an additional paragraph on this 😊

- “Physical limit means we can only hold two pens at a time, cognitive bottleneck means we can only write with one of them at a time”

Bottleneck

- Imagine a motorway with many lanes. Everyone has to pay a toll.
- But the government doesn't have enough \$ to pay for more than one toll booth (this is in a country that doesn't have e-tags).
- The capacity to process tolls is equal to one car at a time.
- Thus everyone has to slow down when they near the toll booth because only one car can be processed at a time.

If you're feeling anxious, remember that in China there's a 50-lane highway that narrows down to just 4
by u/jkitty_1960 in interesting

link

Overt and Covert Attention

6 responses

<>

Covert and overt attention...

overt vs covert attention...

Still about overt VS covert...

did we start attention already??...

sometimes the unconscious functions are made aware (for example im aware of my breathing), suddenly you cant breath normally anymore, but need to "forget" about it so it could return to normal. why?...

obligatory caleb owens mention...

Please read **Chapter 4** and then give me a more specific question 😊

What are you unclear on from today's lecture?



Two-Letter Task

4 responses



What the two-letter experiment proves ...

Whether the two letter task is actually a good paradigm for attention, or whether it's just a visual memory thing, or how those relate ...

the letter thing ...

why it is possible for us to read at high speed (think speed reading tests) but can only process one letter at a time ...

Suggests limited processing capacity. For another task, see Ch. 3.4

Learning Outcomes Lecture 2

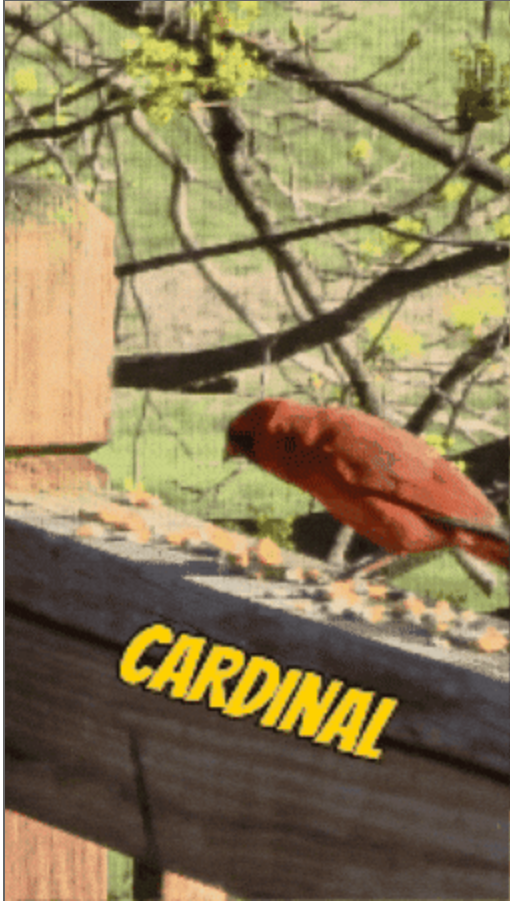
- **Understand vigilance** (briefly mentioned in Ch. 5, 16)
 - Define *vigilance*'s ecological purpose (monitoring the environment for threats and predators)
 - Explain the “nocturnal bottleneck” in evolution and why brains require an interrupt mechanism
- **Distinguish concentration from vigilance**
 - Describe how the nervous system balances goal-directed focus (**concentration**) against environmental alarms (**vigilance**) and the relationship to bottom-up vs. top-down attention
 - Contrast this biological interrupt architecture with digital computer interrupt signals
- **Distinguish bottom-up vs. top-down attention** (Ch. 5)
 - Define *bottom-up* (*exogenous / stimulus-driven*) attention: rapid, involuntary, triggered by sensory salience
 - Define *top-down* (*endogenous / goal-directed*) attention: voluntary, slower, driven by intentions, tasks, memory

Vigilance

- For behavioural ecologists, *vigilance* is monitoring surroundings for predators
- Hard to get close. Why? Vigilance
- The arrival of predators is an example of something that may be more important than whatever an animal is doing

Vigilance

- For ecologists, *vigilance* is monitoring surroundings for predators



Vigilance - define its ecological purpose



- Because of predators many animals (including most mammals!) are nocturnal, and vigilant

Our nocturnal ancestors

- Prior to the dinosaurs, our ancestors had vision specialized for daytime.

Our nocturnal ancestors

- During the dinosaurs, mammalian ancestors shifted into the night



Our mind evolved in a predatory environment



Vigilance - Explain the “nocturnal bottleneck” in evolution and its relationship to vigilance

- Vigilance - periodically look/listen/feel/smell around
- Balance focus against watching out for more important things appearing

Vigilance in deer, and monkeys





player
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ion error

Error 153

Error 153

Concentration vs. vigilance

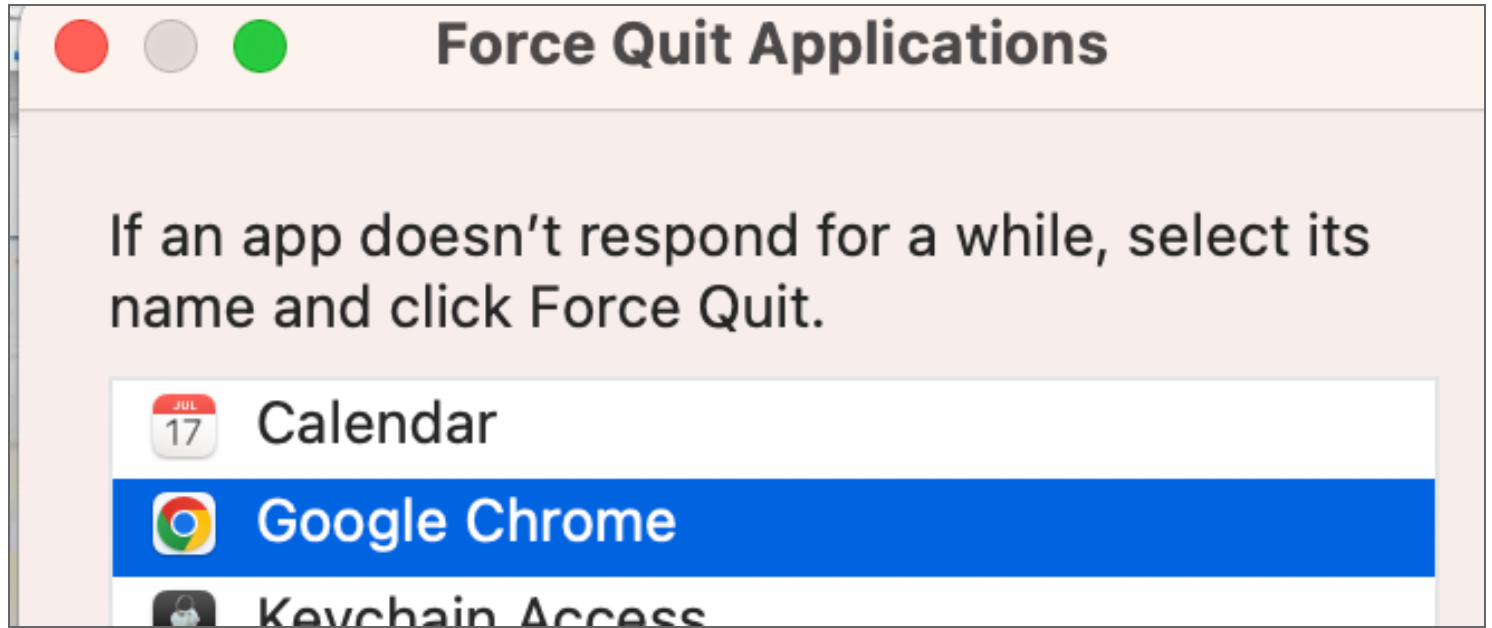
Distinguish concentration from vigilance, balance, relationship to digital computer interrupt signals

- Solution 1: Alternate between concentrating and looking around
- Solution 2: Have a system for overcoming concentration with potentially-important signals

Computers were designed to focus

Computers were designed to focus

- Computer scientists had to add “interrupt signals”
- Keystroke interrupt
 - CTRL-C / Alt-F4 (Windows) / Option-Command-ESC (MacOS)



Animals had to be more interrupt-able, to not get eaten

Animals had to be more interrupt-able, to not get eaten

- Humans don't freeze - We have more robust interrupts than computers

Bottom-up / exogenous attention

Bottom-up / exogenous attention

:

Bottom-up attention

- “Salient” stimuli
- What is salient to us?
 - Large changes to a basic feature of the scene
 - Different than nearby stimuli in terms of low-level features, see Chapters 7 and 9
- Is a kind of vigilance
 - Basic alerting system
 - Like a smoke alarm: Dumb, but requires few resources. Cheap to run it all the time.



Bottom-up attention

Distinguish concentration from vigilance, balance, relation of bottom-up attention to digital computer interrupt signals

Top-down attention

- Sometimes called *endogenous* attention
- Internally driven
- Goal directed

What most often distracts during a lecture?

- Bottom-up vs. top-down attention

Understanding the hand-wave and monster onset

- My hand motion triggered a shift of attention to it. How?
 - Visual cortex signals the location of odd motions
 - Bottom-up attention encourages a shift of attention
- Your attention may leave my hand quickly, because it's not interesting
- The monster appeared on the screen.
 - Visual cortex signalled the location of a sudden onset (Ch. 6)
 - Bottom-up attention encourages a shift of attention
- Attention may linger, because the monster is interesting (top-down attention)

Bottom-up attention

1. The monster appeared on the screen
 1. Visual cortex signalled the location of a sudden onset (Ch. 6)
 2. Bottom-up attention encourages a shift of attention
 3. *Only then* is the monster routed through the bottleneck and recognized to be a monster
 4. Attention may linger there, because monsters are interesting / arousing
- Steps 2-3 happen faster than you can notice!

Did attention go to the monster because it was a monster? NO!

- The monster appeared on the screen
- Visual cortex signalled the location of a sudden onset (Ch. 6)
- Bottom-up attention encourages a shift of attention
- *Only then* is the monster routed through the bottleneck and recognized to be a monster.
- Attention may linger there, because monsters are interesting / arousing

Distinguish concentration from vigilance, balance, relation of bottom-up attention to digital computer interrupt signals

Distinguish bottom-up from top-down attention

- Distinguish bottom-up vs. top-down attention (Ch. 5)
- Define *bottom-up* (*exogenous / stimulus-driven*) attention: rapid, involuntary, triggered by sensory salience
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Distraction experiment

- Put your phone on airplane mode.
- Turn off notifications on your computer.
- Go to phone stopwatch (in “Clock” on iPhone and most Android phones).
- Press “lap” each time:
 - you can’t help yourself and you check social media,
 - think about something other than what you’re reading
 - otherwise get off task

Start stopwatch and silently read Ch. 5, and Ch. 7 if you finish 5

Chapter 5

Press 'lap' every time you get distracted.

8511 9832

or <https://www.menti.com/al91wfyzbom>



PSYC2016: Attention

Speaker notes